

A Bibliometric and Cluster Analysis of Blended Learning Literature

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Abstract—The objective of this document is to analyze the evolution of the scientific literature on blended learning in the university context over the past 10 years by using bibliometric analysis and cluster analysis. Different descriptive statistical techniques commonly used in bibliometric analysis and science mapping were used to analyze the bibliographic dataset; in addition, we use co-word analysis and network analysis to examine author keywords and search for thematic clusters. Thus, six thematic groups were identified. Most of these groups focus on qualitative analysis of the experiences of hybrid courses. Other clusters focus on the technological adaptations of the courses and on evaluating the results of these adaptations from the perspective of the students. Few documents contain reports related to the design, development, implementation, and evaluation of a BL course in higher education with a teaching approach.

Index Terms—Blended learning, higher education, bibliometric analysis.

I. INTRODUCTION

THIS article presents an analysis of the evolution of the main documents, authors and emerging terms in the field of Blended Learning (BL). Didactic proposals in university education focus on building mixed curricular structures, i.e., face-to-face sessions (traditional model) and virtual activities (resources) with the purpose of strengthening the learning experience through a flexible, dynamic, and autonomous process. This combination of face-to-face and virtual teaching methods mediated by technological resources—known by the generic term Blended Learning [1], [2]—is a clear trend in the context of higher education in the last decade [3], [4], [5], [6], [7]. Interest in its adoption is due, among other things, to its ability to serve more students, facilitate access to the academy by replacing some face-to-face sessions with virtual meetings or virtual activities, and improve learning by adopting and adapting various technological resources and digital devices [3], [8]. The BL concept states in theory a formative potential that contrasts with practical evidence. The coherent integration of information and communication technologies in virtual

environments and face-to-face scenarios is more complex than adding digital resources to the classroom [9], [10]. In fact, combining technological tools and different learning environments is not something new; teachers constantly combine pencil, paper, marker, board, slide projectors, video players and many other educational resources within the classroom, in a specialized laboratory, or in an outdoor space [11], [12], [13].

Academic reports on BL are aimed at describing practical implementations of combined curricular structures, present practical interpretations of the generic concept of BL [14], [15], [16], [17], identify key factors that help determine success [18], [19], [20], and identify problems related to perception [21], performance [22], [23] and student affinity in this form of blended teaching-learning [24], [25]. However, these reports form a large body of literature that brings together various approaches to BL, which until now has not been fully mapped by previous works. Therefore, it is necessary to structure, organize, and classify the different contributions and determine which are the thematic areas into which current publications can be divided. Consequently, this research aims to answer the following two research questions: What are the subject areas in which different approaches to BL in higher education can be classified? and what are the characteristics of these thematic groups? The objective of this document is to analyze the evolution of the scientific literature on blended learning in the university context over the last 10 years by using bibliometric analysis. Used metrics include the most cited articles, journals, authors, institutions, countries and keywords, their analysis over time, likewise, the analysis of the thematic groups obtained from the word co-occurrence matrix is presented.

The rest of this document is organized as follows: Section II presents a definition of terms and concepts on Teaching Analytics and Learning Analytics. Section III describes the research protocol. In Section IV, the results obtained are analyzed and discussed. And finally, in Section V, the research question posed is answered.

II. RESEARCH PROTOCOL

The following workflow is applied in this document, which has been adapted from the scientific mapping methodology, systematic literature analysis, and bibliometric analysis [26], [27], [28], [29], [30], [31], [32], [33]:

- Data search, collection, and cleaning.
- Data analysis.
- Data visualization.
- Data interpretation.

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A. Data Collection, Extraction, and Cleaning

For this research, the Scopus bibliographic database was used. The search was conducted in July 2020 and included documents published in the last 10 years, from January 2010 to July 2020. The following search string was used to obtain the documents used for the analysis:

(TITLE (“blended learn*” AND (“higher education” OR university)) OR (KEY (“blended learn*” AND (“higher education” OR university)))) AND (EXCLUDE (PUBYEAR, 2009) OR EXCLUDE (PUBYEAR, 2008) OR EXCLUDE (PUBYEAR, 2007) OR EXCLUDE (PUBYEAR, 2006) OR EXCLUDE (PUBYEAR, 2005) OR EXCLUDE (PUBYEAR, 2004) OR EXCLUDE (PUBYEAR, 2003) OR EXCLUDE (PUBYEAR, 2002) OR EXCLUDE (PUBYEAR, 1984))

Since the term BL appears in last decade, only documents published from 2010 onwards were used. Due to the large number of items recovered, an additional exclusionary criterion was not applied to the year of publication of the documents. The information collected was as follows:

- Article title.
- Authors.
- Author’s Scopus ID.
- Author’s affiliation.
- Source title.
- Abstract.
- Publishing year.
- Author keywords.
- Index keywords.

For the preparation of this document, text processing techniques and manual review were applied to clean the obtained bibliographic data set. The tasks performed were:

- Author name normalization.
- Author name disambiguation: we search for identical author names with different Scopus ID.
- Identification of the country and institution names from the Affiliation field.
- Extraction of keywords from the title and abstract.
- Text unifying for keywords differing in British and American spelling, and the use of singular or plural form.
- Manual review of extracted country and institution names.
- Manual grouping of words that differ in writing but have the same meaning or point to the same concept

B. Data Analysis, Visualization and Interpretation

Different descriptive statistical techniques commonly used in bibliometric analysis and science mapping were used to analyze the bibliographic dataset; in addition, we use co-word analysis and network analysis to examine author keywords and search for thematic clusters.

III. OBTAINED RESULTS

A. Descriptive Statistics

The search was conducted on July 30, 2020 and a total of 939 documents were automatically retrieved. The documents were published in 485 different sources (430 articles, 2 articles in press, 3 books, 70 book chapters, 421 conference paper

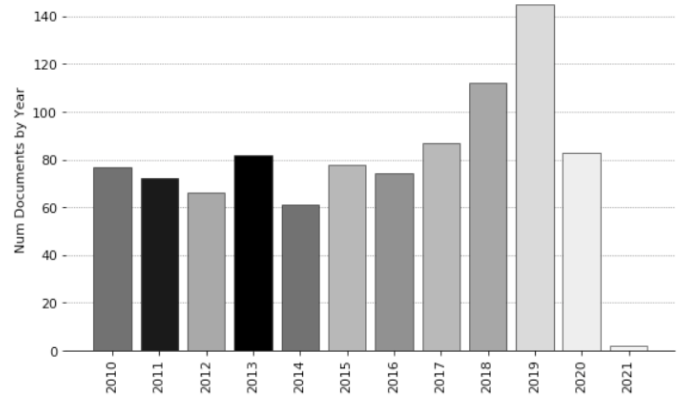


Fig. 1. Number of publications per year. The intensity of color of the bars is proportional to the number of total citations of the documents published in the corresponding year.

and 13 review) and written by 2196 authors affiliated with 893 institutions in 91 countries. There are 736 multi-authored documents and 202 single-authored documents (there is one article without authors), with a mean of 2.34 authors per document and 0.43 documents per author; for multi-authored documents, there are an average of 2.73 co-authors per document. The authors cite on average 28 references per document. Retrieved documents have 1,969 different author keywords and 2,761 index keywords. After the review described in the methodology (text unification and manual review), the author and index keywords were reduced to 1,706 and 2,616 respectively. It is necessary to notice that there are documents published after the search was carried out, which were considered in this analysis.

B. Analysis by Year

This subsection presents the publication results and citation averages per year for the last 10 years. Fig. 1 shows the number of documents published by year; the maximum occurs in 2019 with a total of 145 documents published, and at the time of writing this report there are two documents published for 2021 of the authors Thabet et al. [9] and Kousar et al. [34].

Thabet et al. [9] report the perceptions of teachers regarding the implementation of BL based on the integration of information and communication technologies (ICTs) and the learning management system (LMS) platforms at the tertiary level of higher education. This report concludes that the great barrier that inhibits the adoption of technology in a combined model is the lack of a clear integration policy. On the other hand, Kousar et al. [34] propose an instructional design model based on the design of the Situated Cognitive Processing (SCP) model for a combined structure course that focuses on the student from the instructor’s perspective.

The graph shows stability between 2009 and 2016 with an increasing trend since 2017, indicating a growing interest of researchers in this area. In addition, the intensity of color of the bars in Fig. 1, is proportional to the total citations of the documents published in each year.

In Fig. 2, the number of total citations by year is presented. Notice that, the height of the bars is proportional of the

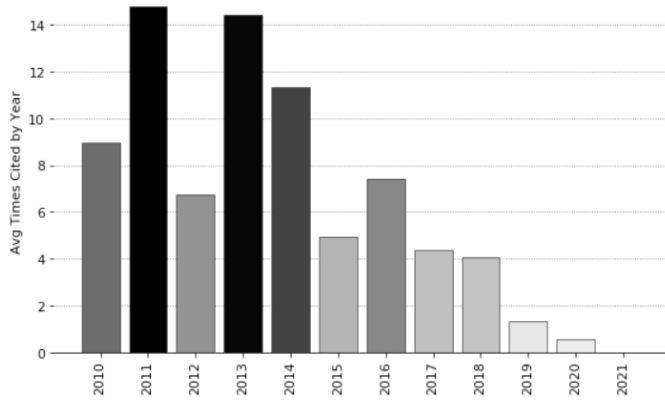


Fig. 2. Average times cited per year. Intensity of color is proportional to the height of the bar.

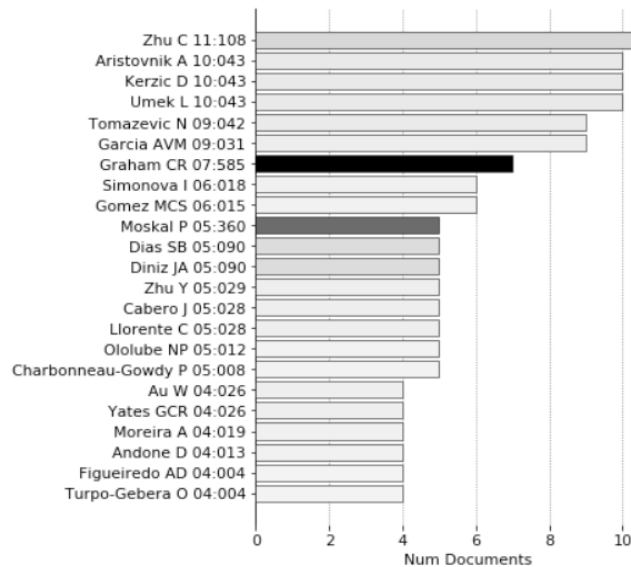


Fig. 3. Number of documents published by author, for authors with four or more documents in the dataset. Intensity of color of the bars are proportional to the number of total citations. Numbers following authors correspond to the number of documents and total citations.

intensity of color of the vertical bars in Fig. 1. The decreasing trend that has occurred since 2016 corresponds to normal behavior since more recent articles have fewer citations than older articles. This graph presents several points of local maxima in citations for the years 2011, 2013 and 2016. The causes of this behavior are discussed below.

C. Analysis by Term

Fig. 3 presents the authors with four or more documents; numbers following the author's name correspond to the number of documents published by the author and the number of citations to the author. There are 23 authors listed in the graph. The intensity of color of the bars is proportional to the number of documents. As a complement, Fig. 4 presents the authors with 100 or more citations; the intensity of color of bars in Fig. 4 is proportional to the number of documents published by the author. In addition, Table I presents the most cited documents.

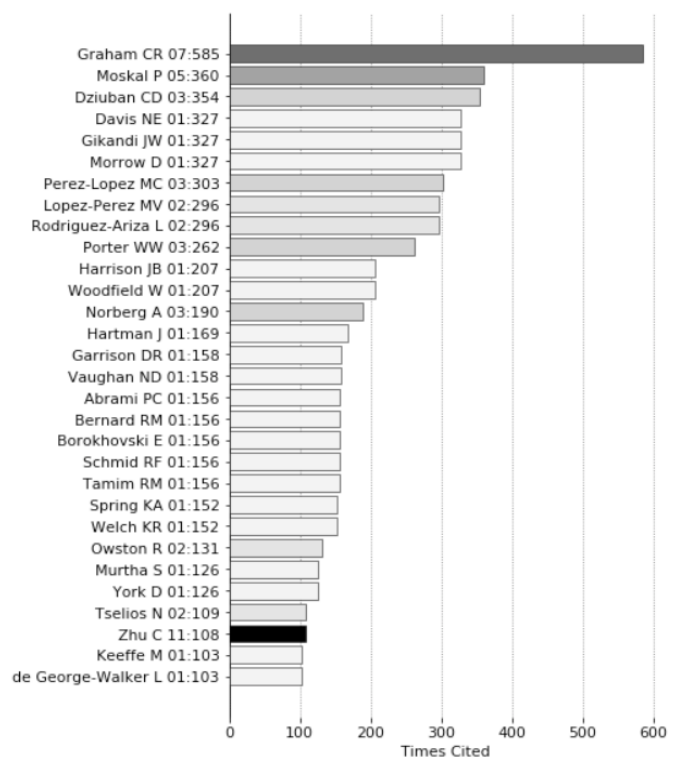


Fig. 4. Citations by author, for authors with 100 or more citations. The intensity of color of the bars is proportional to the number of total citations. Numbers following the author's name correspond to the number of documents and total citations, respectively.

The inspection of Figs. 3 and 4, and Table I, reveals that only authors Zhu C, Graham CR, and Moskal P, simultaneously appear in them, which indicates a permanent and high-impact research in the area; in fact, Graham CR is the author of the third and seven most cited papers in Table I, while Moskal P is the author of the fourth and eighth most cited articles in Table I. In the same sense, many of the authors that appear in Fig. 3 are authors of the documents listed in Table I. Moreover, the years of publication of these documents explain the maximum values for the years 2011 and 2013 in Fig. 2.

The work of Zhu C is primarily aimed at the study of BL implementations in higher education from the student's perspective [17], [22], [23], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], identifying issues such as motivation, satisfaction, performance, and learner perceptions. In recent years, Zhu C has included teaching experiences in BL adoption in his research [48], [49], [50].

In [51] and [52], Graham CR provides a framework on the levels of adoption and implementation of BL curriculum structures in higher education. His works identify three categories: strategy, structure, and support, which in turn differentiate three stages of adoption to show how institutions move from the adoption of incipient practices in BL toward a mature institutionalization of hybrid dynamics.

Moskal has focused his research on understanding the BL model in higher education in greater detail [53], [54], so it is recurrent to find that his reports comment on the implications of adopting and adapting blended curricular structures

TABLE I
PUBLISHED DOCUMENTS CITED BY 100 OR MORE DOCUMENTS

AUTOR. YEAR. DOCUMENT TITLE	CITATIONS
GIKANDI JW, MORROW D, DAVIS NE. 2011. ONLINE FORMATIVE ASSESSMENT IN HIGHER EDUCATION: A REVIEW OF THE LITERATURE [56]	327
LOPEZ-PEREZ MV, PEREZ-LOPEZ MC, RODRIGUEZ-ARIZA L. 2011. BLENDED LEARNING IN HIGHER EDUCATION: STUDENTS' PERCEPTIONS AND THEIR RELATION TO OUTCOMES [57]	290
GRAHAM CR, WOODFIELD W, HARRISON JB. 2013. A FRAMEWORK FOR INSTITUTIONAL ADOPTION AND IMPLEMENTATION OF BLENDED LEARNING IN HIGHER EDUCATION [52]	207
MOSKAL P, DZIUBAN CD, HARTMAN J. 2013. BLENDED LEARNING: A DANGEROUS IDEA? [53]	169
GARRISON DR, VAUGHAN ND. 2012. BLENDED LEARNING IN HIGHER EDUCATION: FRAMEWORK, PRINCIPLES, AND GUIDELINES [58]	158
BERNARD RM, BOROKHOVSKI E, SCHMID RF, TAMIM RM, ABRAMI PC. 2014. A META-ANALYSIS OF BLENDED LEARNING AND TECHNOLOGY USE IN HIGHER EDUCATION: FROM THE GENERAL TO THE APPLIED [59]	156
PORTER WW, GRAHAM CR, SPRING KA, WELCH KR. 2014. BLENDED LEARNING IN HIGHER EDUCATION: INSTITUTIONAL ADOPTION AND IMPLEMENTATION [51]	152
OWSTON R, YORK D, MURTHA S. 2013. STUDENT PERCEPTIONS AND ACHIEVEMENT IN A UNIVERSITY BLENDED LEARNING STRATEGIC INITIATIVE [60]	126
NORBERG A, DZIUBAN CD, MOSKAL P. 2011. A TIME-BASED BLENDED LEARNING MODEL [54]	121
DE GEORGE-WALKER L, KEEFFE M. 2010. SELF-DETERMINED BLENDED LEARNING: A CASE STUDY OF BLENDED LEARNING DESIGN [54]	103

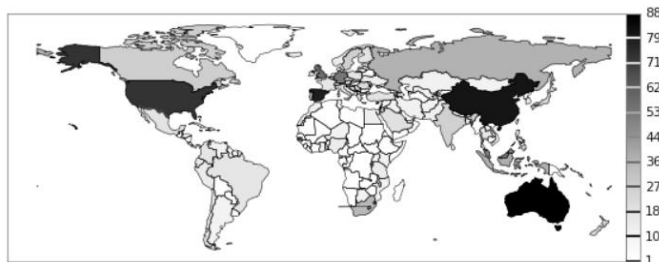


Fig. 5. Worldmap with number of documents published by country. The color intensity of the right-side bar is proportional to the total number of documents published.

in universities as well as recognizing factors in blended learning that may suggest positive institutional transformation [4], [41], [55].

Fig. 5 shows a world map indicating the number of documents published by country. The most frequent countries are Australia, China, Spain, United States, and United Kingdom with 88, 81, 78, 72, and 52 documents. All of them, are in the group of ten most cited countries. Fig. 6 presents the number of documents published by co-authors affiliated with institutions in the same country (SD – Single-authored Document) and co-authors affiliated with institutions from different countries (MD – Multi-authored Document), for the ten countries with the highest number of documents; as can see in the graph, authors prefer to work with authors in the same country

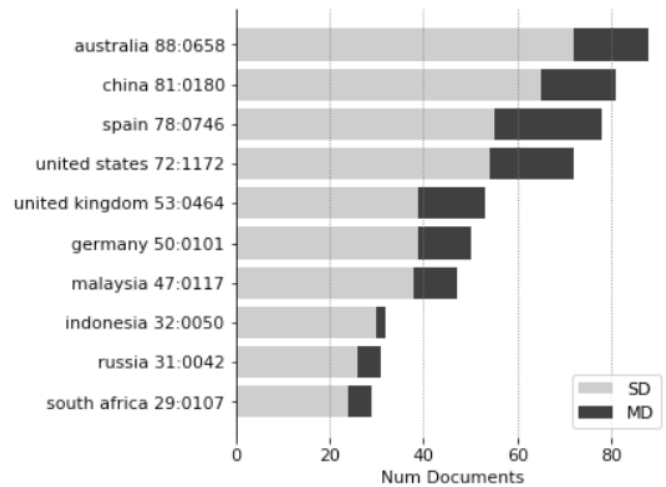


Fig. 6. Number of multi-authored (MD) and single-authored documents (SD) per country.

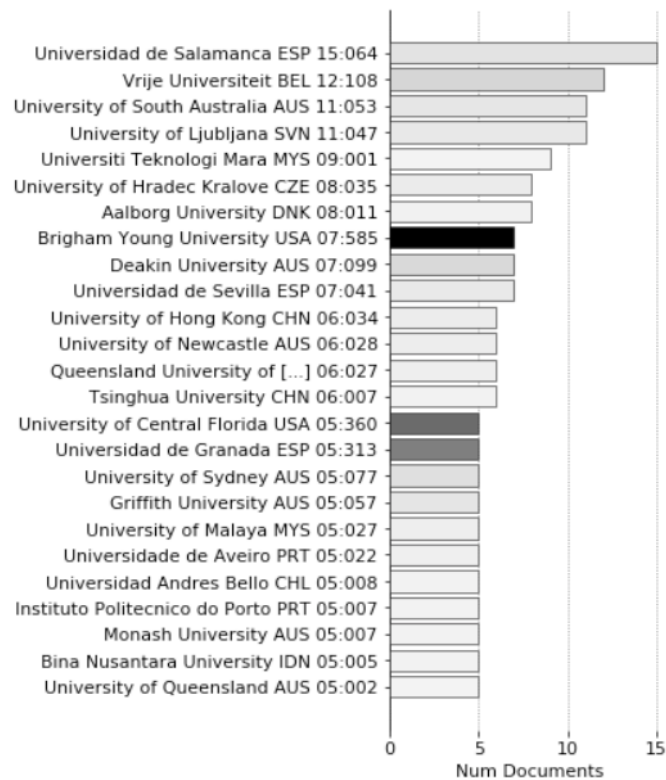


Fig. 7. Institutions with five or more documents published. The darkness of bars are proportional to the number of total quotations. Following numbers the abbreviation for the country of affiliation of the institution corresponds to the number of documents published and the total citations, respectively.

(low international collaboration), with Indonesia being the country with lowest international collaboration.

In addition, Fig. 7 presents the institutions (with their corresponding country of affiliation), with five or more documents. Australia participates with seven universities (University of South Australia, Deakin University, Queensland University of Technology, University of Newcastle, University of Sydney, Griffin University, and Monash University), China with two institutions (University of Hong Kong and Tsinghua University); Spain with three (Universidad de Salamanca,

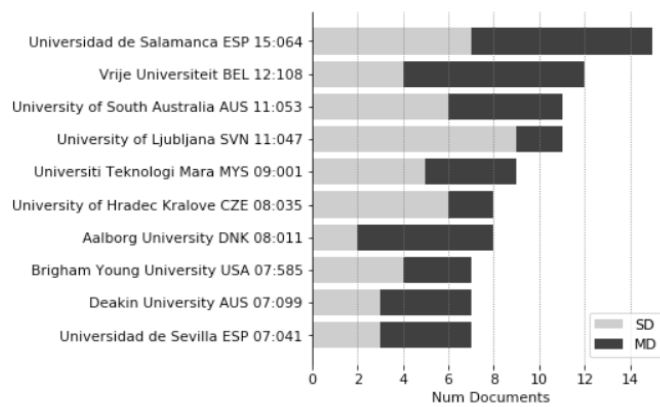


Fig. 8. Number of multi-authored (MD) and single-authored documents (SD) per institution.

Universidad de Sevilla, Universidad de Granada), and United States with two (Brigham University, University of Central Florida).

Fig. 8 presents the number of documents with co-authors affiliated with the same institution versus the number of documents with co-authors at different institutions. As can be seen, most of most documents tend to be written by researchers at different institutions. However, Fig. 6 shows that the institutions of co-authors tend to be located in the same country.

Fig. 9 presents the most frequent keywords in the dataset (with a minimal appearance of 5 times). Keywords used in the search string (as *blended learning*), context-specific keywords (as *English for specific purposes*), or keywords with very vague or imprecise meaning, as *effect*, were excluded from the graph. It is not surprising that the most frequent keyword is *e-learning*, due to the boom it has had in the last decade, particularly with efforts dedicated to the development of online courses both for mass consumption (as a means for professional development) and within organizations.

The titles of sources with seven or more documents in the analyzed dataset are presented in Fig. 10. While Lecture Notes in Computer Science is the journal with most documents published, the most influential source title is Computers and Education with nine documents published and 1,069 citations. The paper with the most citations in the Computers and Education journal corresponds to “Online formative evaluation in higher education: a literature review” authored by Gikandi et al. [56]; this paper presents a qualitative analysis of key issues such as techniques, tools and key actors around formative assessment in virtual and mixed environments in the context of higher education, using the methodology of systematic literature review. In addition, the group made up for the ten most cited source titles has a at least 100 citations; however, six of them have three or fewer documents published.

D. Analysis of Terms by Year

Fig. 11 shows the number of documents published per year for authors with four or more documents in the dataset. This graph shows the publishing pattern for the selected authors; as can be seen, Aristovnik [15], [61], [62], [63], [64], Kerzic

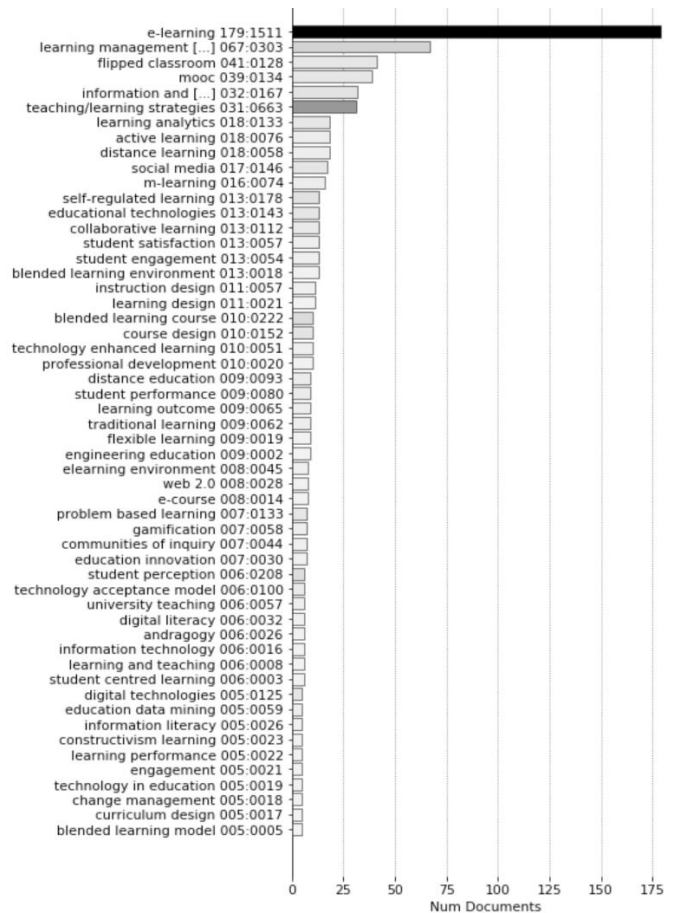


Fig. 9. Author keywords. Following numbers the keyword correspond to the number of documents in which the term is included as a keyword and the total number of citations of these same documents, respectively.

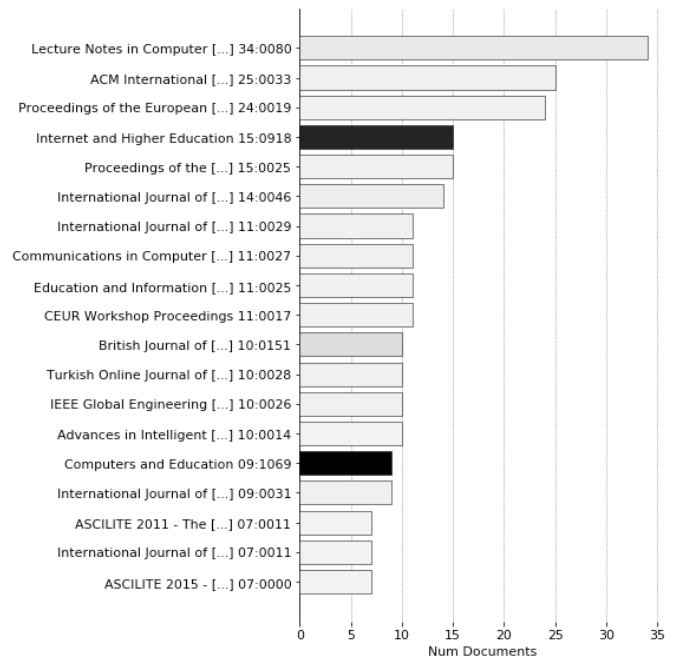


Fig. 10. Number of documents published by source title. The intensity of color of the bars is proportional to the number of citations to the source.

[20], [65], Umek and Tomazevic [66], [67], [68], [69], [70], present the same publishing pattern, and their research activity

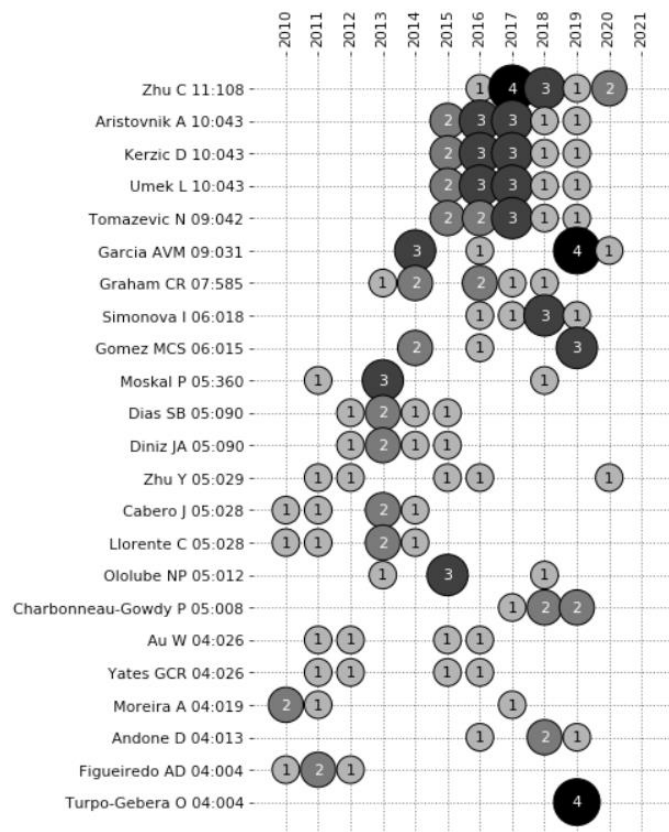


Fig. 11. Number of documents published per author. Numbers following the author's name correspond to the number of documents and total citations, respectively. The numbers within the circles indicate the number of papers published by the author in the year where the circle is located.

is concentrated from 2016 to 2019. The research of this group of authors focuses on the analysis of e-learning and blended learning implementations, from the perspective of learners.

Others authors presenting a period of continuous research are Dias and Diniz [71], [72], [73], Simonova [74], [75], [76], Graham [52], and Figuerido [77], [78], [79], [80]; these works focus on theoretical issues of blended learning, the study of key factors for the adoption and adaptation of the BL proposal at different levels, or the projection of the concept. Another authors have a more intermittent publishing pattern. In dataset, the maximum number of documents published per year is four (Zhu [40], [43], [45], [46], Martin Garcia [81], [82], [83], [84], and Turpo-Gebera [85], [86], [87], [88]).

Fig. 12 presents the number of documents published per year for the most frequent countries. Most countries publish frequently on the topic; but it is notorious that countries as Russia or Indonesia have only published recently. At the time of writing this paper, the last document published by an institution in Russia was “Unlocking students ‘motivation in the blended higher education classroom: Lecturers’ perspectives” by Zavyalova [89]; and the most recent publication with an author's affiliation to Indonesia was published by Wiharja et al. “Student's Perception in Learning Indonesian Language from Conventional to Blended Learning Model” [90]. China has published the maximum number of documents

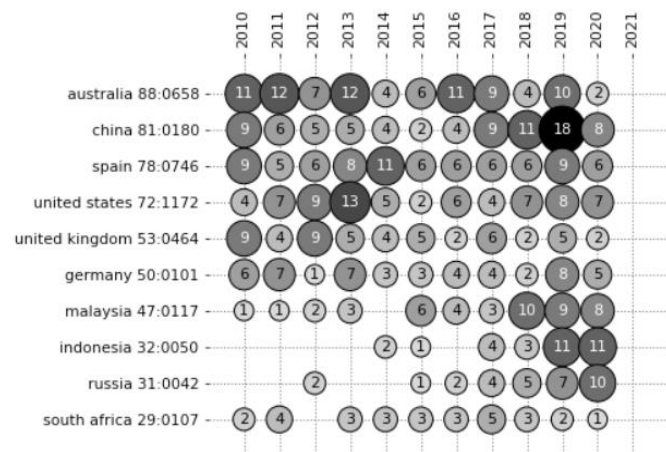


Fig. 12. Number of documents published by country. The numbers within the circles indicate the number of documents published with affiliation to the country and in the corresponding year.

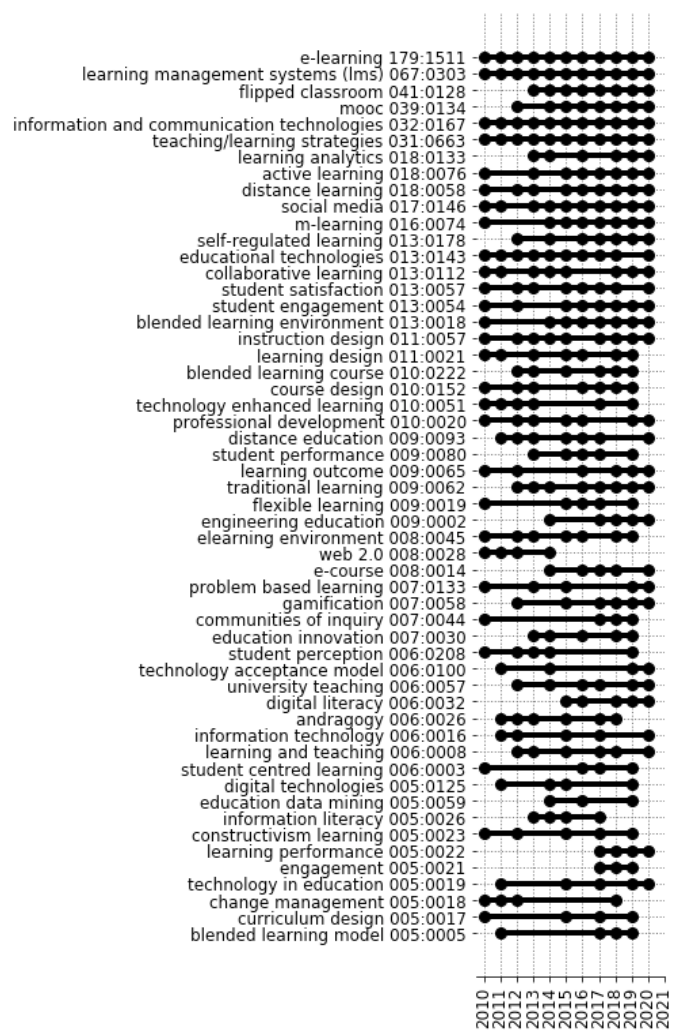


Fig. 13. Gantt diagram for author keywords with an appearance of five or more.

in a year with 18 articles in 2019, followed by United States with 13 articles in 2013.

TABLE II
GROUPS OF THEMES OBTAINED BY CLUSTERING THE MATRIX
OF CO-OCCURRENCES OF AUTHOR KEYWORDS

CLUSTER	KEYWORDS
0	Learning Management System (LMS), m-learning, student satisfaction, Self-regulated learning, engineering education, gamification, e-learning environment, education innovation, digital literacies, information technology, learning performance, technology in education, curriculum design.
1	E-learning, blended learning course, instruction design, learning outcome, communities of inquiry, technology acceptance model, traditional learning, andragogy, education data mining, change management.
2	Mooc, information and communication technology, teaching/learning strategies, distance learning, professional development, distance education, flexible learning, e-learning course, digital technologies.
3	Student engagement, blended learning environment, student performance, problem-based learning, flipped learning, university teaching, learning and teaching, student centered learning, constructivism learning.
4	Flipped classroom, learning analytics, active learning, education technology, learning design, technology enhanced learning, information literacy.
5	Social media, collaborative learning, course design, web 2.0, student perception

Fig. 13 presents the Gantt diagram of the author's keywords used in Fig. 9. This graph shows the time periods in which a keyword has been used. It is remarkably interesting to note that the keywords: flipped classroom, learning analytics, student performance, engineering education, and digital literacy appear from 2013 onward.

IV. DISCUSSION

This section presents the answer to the two research questions raised in this document: What are the subject areas in which different approaches to BL in higher education can be classified? and what are the characteristics of these thematic groups? To answer these questions, we built the author's keyword co-occurrence matrix [91] (normalized with the association index) and was clustered using Louvaine's algorithm [92]; the detected clusters are presented in Table II.

The co-occurrence matrix is a square matrix that indicates the number of times a row word is associated with a column word. The diagonal is the number of times the word appears in all documents. The association index is a form of standardization to highlight relationships between data. For example, knowing when two words appear together beyond the probability that they appear together as separate words. This index emphasizes what is beyond probability. Since this co-occurrence matrix can be interpreted as a social network, Louvaine's clustering algorithm was used to recognize the nodes of the matrix that tend to appear together to do the clustering. We identified and analyzed six clusters related to: a) educational platforms implemented in BL models; b) frameworks to design a BL course; c) comparisons on the performance of students in face-to-face, blended learning and e-learning courses; d) experiences on the adoption and adaptation of BL strategies at the institutional level; e) experiences of

using technological tools and digital resources in BL courses; and f) perceptions and implications on the increased use of technology in combined teaching and learning processes. The rest of this section is aimed to analyze the obtained clusters.

A. Thematic Description

The thematic description of the clusters was made from the analysis of the 20 most cited texts in each grouping, and the analysis of the local meaning of keywords that make up each cluster obtained when analyzing these documents. The documents selected in each cluster were ordered by their number of citations to make their description; in addition, priority is given to documents occupying the first three positions, as well as those with a prominent perspective among the group.

1) *Cluster 0*: This cluster is the smallest of the six clusters identified. It consists of only two documents, the research by Vázquez-Martínez and Alducin-Ochoa [93] on educational platforms in BL models and the work of Ravishankar and Jones [94] which is focused on the design of the online components for an advanced level course in engineering.

Vázquez-Martínez and Alducin-Ochoa [93] conducted a case study in a Spanish university to evaluate learning approaches in a course with BL methodology as a strategy to mitigate student dropout and low academic performance. There are two defined platform learning approaches defined in relation to WebCT-mediated interactions: superficial and deep. The superficial approach is related to the students who do not perceive a value relationship with the assigned academic tasks and therefore do show no greater interest in doing them. The second approach is totally the opposite since students are interested in tasks, enjoy them, and give them a meaning related to their personal experiences and the professional world. This work concludes that learning approaches influence the positive perception of the use of LMS platforms and student learning outcomes. Students identified with deep focus made best use of the tools presented in blended environments. In addition, Vázquez-Martínez and Alducin-Ochoa emphasize that it is important to understand how each student learns to correctly define the instructional and didactic design of the course.

Ravishankar and Jones [94] presents a case study of an engineering course at a university in Sydney. This research aims to recognize the factors that allow creating meaningful learning experiences in virtual environments for blended modality courses. The experience is based on the CoI (Community of Inquiry) model; an online platform for course delivery is used to integrate the social, cognitive and teacher dimensions of the CoI model. The design of the course was carried out with total rigor. A variety of academic resources and some informal, popular culture digital resources were included to bring students closer together. The design of the course also considered every detail in relation to the interaction of students and teachers on the platform. This work highlights that the CoI model implemented along with all the detailed configuration deployment of the course were crucial to achieve positive learning outcomes and teacher performance.

2) *Cluster 1*: The documents that are part of this cluster are focused on three aspects; first, in analyzing specific e-learning

interactions to take full advantage of virtual blended learning environments [95], [96]; second, identify the key factors for the transition to a BL model; and third, analyze the relevance of instructional design to achieve the benefits claimed by the combined model. The research conducted by Adekola et al. [97], Radovan and Kristl [98], and Mayisela [99] are the most cited of this group. Adekola et al. [97] aimed to develop a reference framework for the transition to a blended learning strategy at the institutional level. This framework is concerned about stakeholder issues, institutional infrastructure considerations, and organizational readiness for investment.

Radovan and Kristl [98] investigate the acceptance and use of learning management systems (LMS) among the teaching community in a Slovenian university. The authors highlight the cognitive and social complexity that teacher-student interaction can represent in the virtual space; therefore a previous process of teacher training on the use of educational technologies and curricular transformation is necessary before starting a full implementation of a blended learning model, since the perceived ease is higher than the actual ease of use [95].

In [99], Mayisela focuses on the benefits that a combined learning strategy can have on a higher education curriculum. This work identifies that the resulting instructional design is a crucial component to improve student performance, as well as the gradual integration of information and communication technologies and the use LMS platforms facilitates the formative accompaniment of teachers.

3) *Cluster 2*: Cluster 2 gathers documents that investigate the configuration of blended learning strategies and compare them with the face-to-face or e-learning modality [100], [101]. These comparisons are made mainly on the perceptions that students have about BL in relation to their experiences in face-to-face or distance contexts [35], [102]. The investigations of this group also contrast the academic performance of the students, and analyze the results of the evaluation [103] and the impact on the achievement of the learning objectives. Finally, these comparisons join the group of reports that identify key factors for the success of the implementation of a BL course. These success factors are related to digital literacy for students and teachers, the effective use of technology and the strategic design of BL at the institutional level [104], [105].

The most cited documents of the cluster are those of Al-Qahtani and Higgins [106], Tselios et al. and Donnelly [107]. Al-Qahtani's research [106] compares the academic performance of students in face-to-face, blended and e-learning environments. The results show that the performance in the blended learning model was higher than the achieved in the face-to-face and online modes. However, there was no significant difference between performance results in face-to-face and e-learning modalities.

In [108], Tselios et al. identifies the effects of blended learning on the training of students. Tselios et al. [108] takes as reference a case study at a Greek university and makes use of the TAM model to compare the perceptions of students between a distance learning strategy and a blended learning strategy. Tselios' research concludes that there is not a considerable difference in students' perception and behaviors regarding technologies between these strategies.

Donnelly [107] conducts research similar to that of Tselios on the impact of technology on blended learning and includes the experiences of teachers in his analysis. Donnelly et al. identifies crucial factors for the design of a BL experience such as the technical infrastructure, the preparation of teaching materials and training of teachers. This work concludes that the development of the aforementioned factors must be integrated with an institutional approach to achieve better impacts on the academic community.

4) *Cluster 3*: Cluster 3 groups documents that report experiences of adoption and adaptation of BL strategies at the institutional level. It also includes other experiences from the students' perspective and some from the role of teachers. These BL strategies are conceptualized from three key elements: 1) the combination of environments where instruction occurs, i.e. virtual or face-to-face; 2) the combination of the interaction time, which can be synchronous or asynchronous [54]; and 3) the combination of technological means such as LMS platforms [73], used to manage teacher-student interaction at different moments of the training process; or MOOCs implemented to strengthen the approach and integration of the educational experience [4], [109], [110].

In relation to experiences from the perspective of the students, the research by Manwaring et al. [111] recognizes the motivational profiles of students [112] and their behavior in the implemented BL model. The Vanslambrouck report [35] is also important because it addresses the relationships between student motivation and the assessment, they make of the mixed curricular structure course they have experienced. The few texts related to research from the teaching perspective highlight the responsibility of the instructors in the configuration of the didactic sequences of the course. The most cited documents in this group correspond to the works of Gikandi et al. [56], Graham et al. [52], and Moskal et al. [53].

Gikandi et al. [56] conducted a systematic review related to the types of evaluative activities in mixed education and e-learning contexts, and their relationship with the learning evidence. This work identifies that a wide variety of techniques for formative assessment are linked to online tools such as questionnaires and discussion forums. Additionally, these online tools are recognized to have benefits such as enhancing student engagement and engagement and developing a learning community.

Graham et al. [52] investigate the institutional characteristics for the adoption and implementation of blended learning strategies. This research is based on six cases of BL adoption in universities. The main contribution of this research is the description of the three stages passed by a higher education institution in implementing a BL strategy: (i) exploration, (ii) early adoption, and (iii) maturity and growth.

In [53], the research work of Moskal et al., like that of Graham et al. [52] it is carried out at the institutional level. Moskal et al. [53] uses as a model the sixteen years of BL experience of a North American university and identifies the key actions that each of the stakeholders within the system must take. These actions include the definition of institutional policies, the development of a reliable and robust

infrastructure to support teachers and students, and permanent self-evaluation.

5) *Cluster 4*: The documents that are part of cluster 4 are research on the types and possibilities of configuration of a blended learning strategy and the technological tools involved. With regard to the latter, research highlights the use of educational platforms [113], [114], [115], or learning management systems (LMS) as support for the BL model, or mobile devices to bring closer and facilitate interactions between teachers and students [116]. In this group, aspects such as the implications of BL on the behavior of students and teachers are also discussed. On the implications are mentioned: the self-regulation of students in the use of technologies and their responsibility in the review of the material provided in the platforms; the curricular transformation that teachers must execute; and, the technological experimentation to which they must be willing [117], [118], [119]. The works of Kurup and Hersey [120], Bachtiar et al. [121], And Kumpu et al. [122] are the most cited of this group.

Kurup and Hersey [120] investigates the relationships between the types of configuration of a BL course and the learning objectives. Kurup and Hersey [120] identify that content related to lower-order objectives has the greatest impact when delivered virtually and can be accessed at any time by students. While the contents and activities that belong to the higher order of learning must be developed in synchronous encounters. Kurup's review concludes that, although a blended learning strategy represents the best of the classroom model and the potential of technology, it is always necessary for teachers to review the curricular structure very well to achieve the defined purposes.

The work of Bachtiar et al. [121] is based on the technology acceptance model - TAM to identify the perceptions and behaviors of students in e-learning environments.

Although the research results of Bachtiar et al. refer to a completely virtual course scheme, the authors suggest that the results they present may also be valid for blended learning course in the virtual component. In contrast to the previous research, Kumpu et al. [122] focuses their work on the economic factors that influence the adoption of blended teaching strategies at the institutional level and contrasts them with the benefits in the achievement of student learning objectives. Kumpu et al. uses three universities located in South Africa, Sweden and Uganda as a case study, and concludes that although the costs of implementing BL are high due to the technological integration and staff training, the benefits on the quality of academic training they are superior in the medium and long term.

6) *Cluster 5*: The studies that are part of cluster 5 focus on the perceptions and consequences of the increased participation of technology in the teaching and learning processes. Among the most important contributions, it stands out that: the online component of a BL strategy requires a structure aligned with the face-to-face component to positively impact the student's experience [123], [124], [125]; blended learning modalities based on different forms of technology integration lead to positive perception and active participation of students [126], [127]; and finally, massive courses - MOOCs can be

improved through the meaningful use of information and communication technologies [128]. The most cited documents of the cluster correspond to the works of Bozalek and Biersteker [129], le Roux and Parry [130] and Fearon et al. [131].

In [129], Bozalek and Biersteker investigates the effects of the participatory approach to learning in BL on students. This approach encouraged the development of critical and reflective skills in students as it encouraged them to ask questions, take a position on situations, and share their perspectives supported by technology.

The work of Le Roux and Parry [130] focuses on the negative consequences arising from the increased involvement of technology in blended learning initiatives. In this study, special attention is paid to study areas where technological artifacts are the subject of interest and the training tool. Le Roux and Parry recognize relevant issues about beliefs, norms and motivators around technology and BL; for example, the statement of social norms around the use of digital devices that can obstruct learning rather than promote it; or the negative correlation between the use of the media at the conference and academic performance, since the high level of use of these devices, mostly without specific purpose, favors the distraction of students.

Fearon et al. [131] conduct their research with a similar approach to Bozalek and Biersteker to identify, among students' opinions, key issues in terms of the perceived benefits of a blended learning environment. This research concludes that students perceive blended learning as useful since it gives them flexibility in their training process and that interaction with digital devices is also part of those preferences, because it motivates them to exchange ideas and work as a team.

V. CONCLUSION

BL is roughly defined as the combination of face-to-face teaching methods with technological resources and virtual activities; BL has become a clear trend in the context of higher education in the last decade as is evidenced by the vast number of published documents. This paper presents the evolution of the scientific literature on blended learning in the university context over the past 10 years. The literature review that we conducted on BL stands out because it uses bibliometric analysis, statistical methods, and scientific mapping tools for the processing of data and their presentation in a quantitative and qualitative way. First, different descriptive statistics and plots are used to present a descriptive analysis of the analyzed documents. Second, co-word analysis and network analysis are used to examine author keywords and search for thematic clusters. With the co-occurrence analysis of terms we identified and analyzed six clusters related to: a) educational platforms implemented in BL models; b) frameworks to design a BL course; c) comparisons on the performance of students in face-to-face, blended learning and e-learning courses; d) experiences on the adoption and adaptation of BL strategies at the institutional level; e) experiences of using technological tools and digital resources in BL courses; and f) perceptions and implications on the increased use of technology in combined teaching and learning processes.

This work is important because it characterizes approaches to Blended Learning by identifying thematic trends and gaps in the concept. It also recognizes asymmetry in research interests around hybrid curriculum structures in higher education. Our co-occurrence analysis of terms associated with the main concept, Blended Learning, recognizes an interest among authors to present their contributions as descriptions and evaluations of their experiences from the perspective of students. Interestingly, thematic correlations have a strong focus on the use of technological and digital resources for education. While the relations of the BL seem to be less with concepts related to competencies of the pedagogical team, strategic planning of the course or the development of digital resources of a BL course from the teaching perspective.

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